

PROJECT INDUSTRY PPG

Recoverable Assets (RA) & Inventory Risk Management (IRM)

Azure Data Pipeline for Inventory Risk Analytics

Team 8

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Problem & Objectives

Key motivation and goals of the proposed Azure-based inventory risk analytics system.

Problem

- Inventory, production, purchase, supplier, and sales data are stored separately.
- Risky materials are difficult to identify manually.
- Stockout issues may affect customer sales orders.
- Management needs clear and timely inventory risk insights.

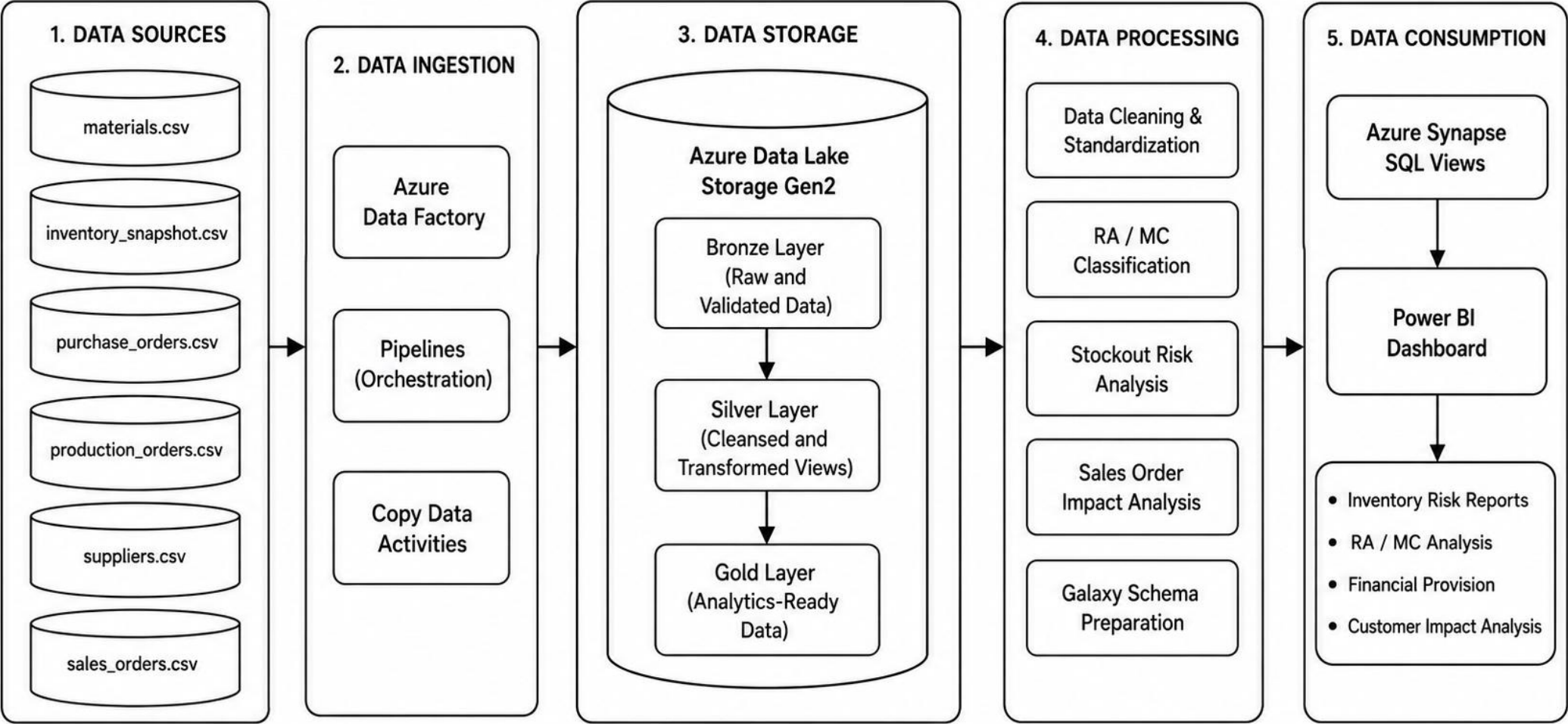
Objective

- Identify Recoverable Assets (RA) and Magna Carta (MC) materials.
- Detect materials below reorder level and at stockout risk.
- Trace impacted sales orders and affected customers.
- Present business-ready insights through Power BI.

Goal: Transform raw CSV data into inventory risk, financial provision, and customer impact insights.

Azure Data Architecture

Logical flow from source data to reporting and dashboard consumption.



Data Sources → Data Ingestion → Data Storage → Data Processing → Data Consumption

System Implementation Flow

Main work completed across Bronze, Silver, and Gold layers.



Key point: The system transforms raw CSV files into cleaned, processed, and dashboard-ready analytical outputs.

Gold Layer Galaxy Schema

Analytical model used to support Q1–Q7 and downstream dashboard reporting.

fact_inventory_status

Supports Q1–Q6

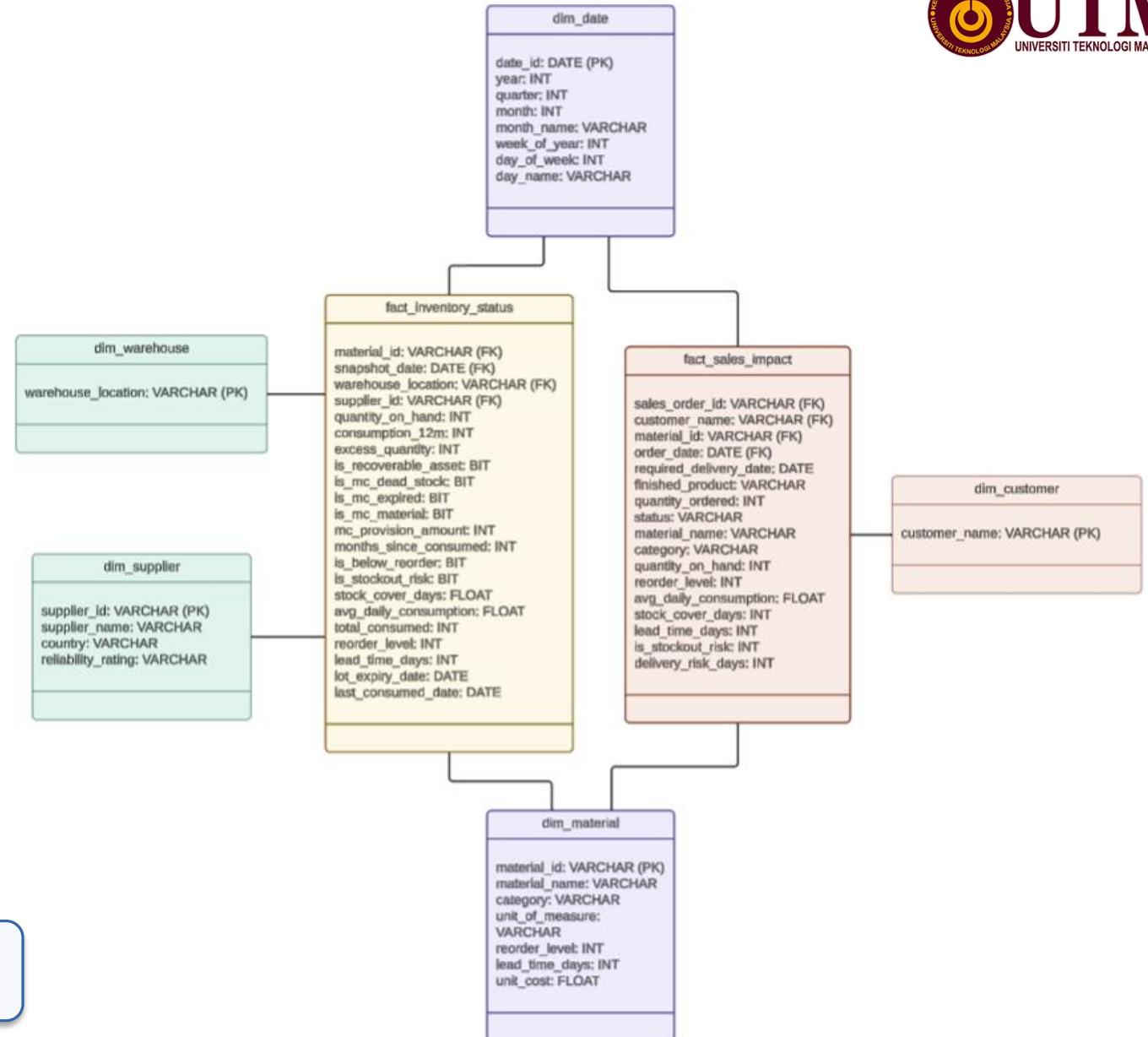
fact_sales_impact

Supports Q7

Shared Dimensions

dim_material
dim_date
dim_warehouse
dim_customer

Why Galaxy Schema? Two fact tables share common dimensions, making the model efficient for analytical reporting.



Final Results

Summary of the main business outputs generated from the implemented logic.

RA / MC Results

Recoverable Assets: 10 materials

MC Dead Stock: 1 material

MC Expired: 2 materials

MC Provision: USD 30,144

Stockout & Sales Impact

Below Reorder Level: 5 materials

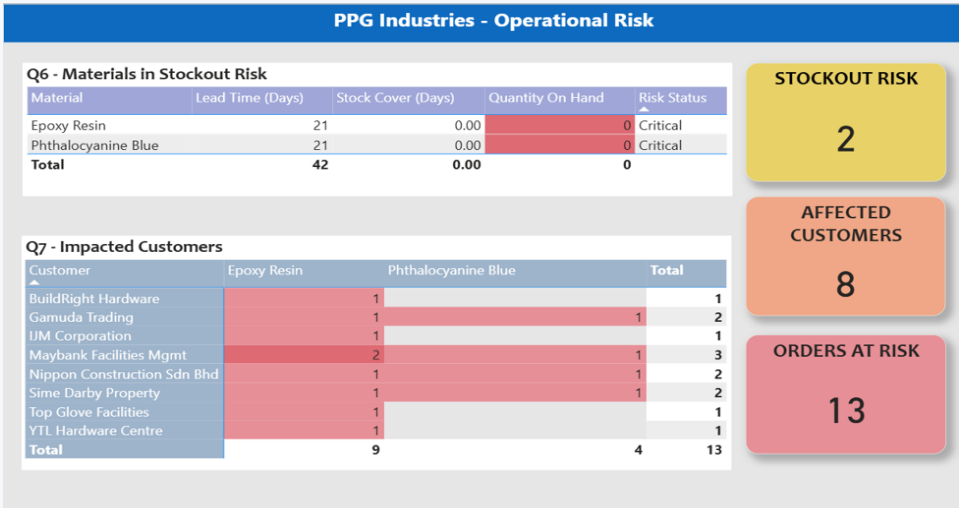
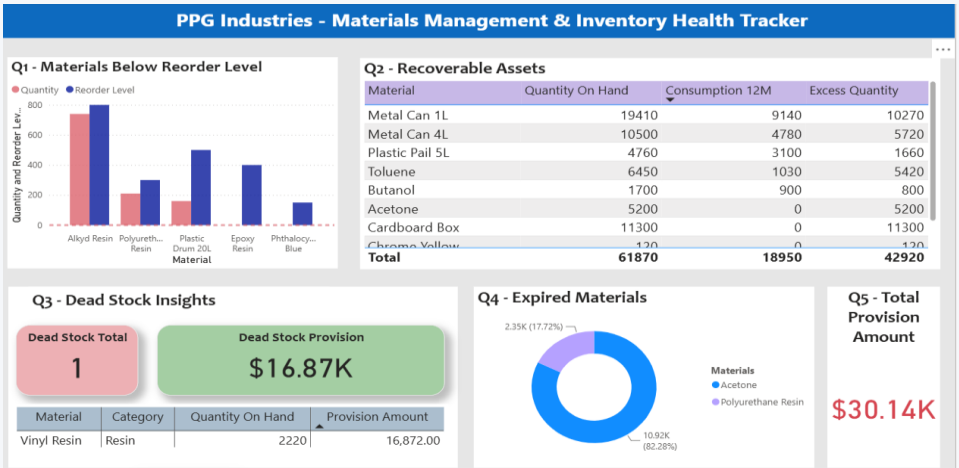
Stockout Risk: 2 materials

Impacted Records: 13

Affected Customers: 8

Business takeaway: The implemented pipeline successfully produced inventory risk, financial provision, and customer impact insights.

Dashboard & Conclusion



Conclusion

This project successfully built an end-to-end inventory analytics pipeline for PPG Industries, from raw data ingestion through to a Power BI dashboard that answers all seven business questions.

The pipeline identified critical stock shortages, recoverable assets, financial provisions totalling \$30.14K, and 13 customer orders at risk of delayed fulfillment. Together, the dashboard equips PPG with a clear, data-driven view of their inventory health and operational risk in one place.

Final message: An end-to-end Azure pipeline was successfully implemented for PPG inventory risk analytics.